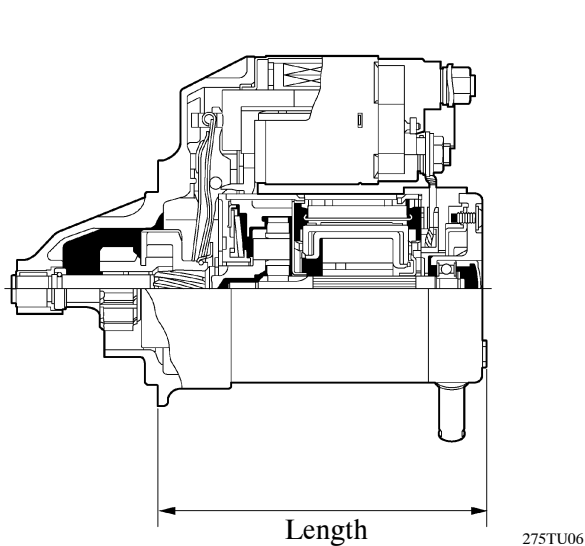


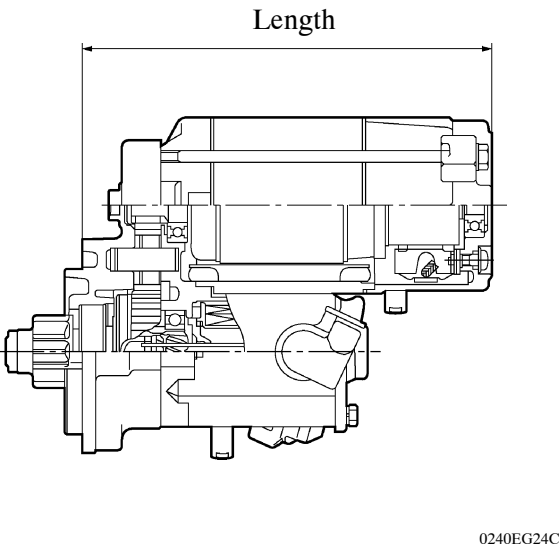
■ STARTING SYSTEM

1. General

- A compact and lightweight PS (Planetary reduction-Segment conductor motor) type starter is used on models for general countries and G.C.C. countries.
- A reduction type starter is used on Chinese models and cold area specification models for general countries.



PS Type Starter



RA Type Starter

► Specifications ◀

Starter Type	PS1.6		RA2.0	
	PS type		Reduction type	
Supplier	DENSO		←	
Rating Output	1.6 kW		2.0 kW	
Rating Voltage	12 V		←	
Length ^{*1}	mm (in.)	126.4 (4.98)	185.3 (7.30)	
Weight	g (lb)	2800 (6.2)	4700 (10.4)	
Rotating of Direction ^{*2}	Clockwise		←	

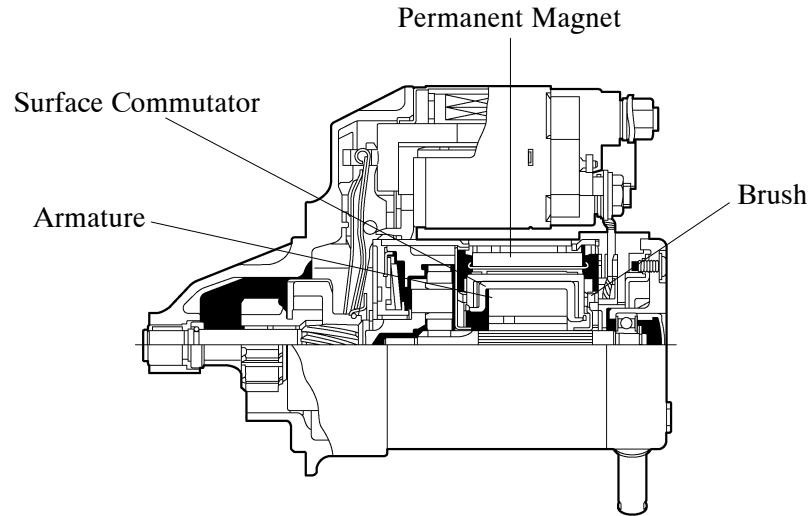
*1: Length from the mounted area to the rear end of the starter

*2: Viewed from Pinion Side

2. PS (Planetary reduction-Segment conductor motor) type Starter

General

- The PS starter contains an armature that uses square-shaped conductors. The surface of the armature at one end functions as the commutator, resulting in improved output torque and overall length reduction.
- In place of the field coil used in the conventional starter, the PS starter uses two types of permanent magnets: main magnets and interpolar magnets. The main magnets and interpolar magnets have been efficiently arranged to increase the magnetic flux and to shorten the length of the yoke.

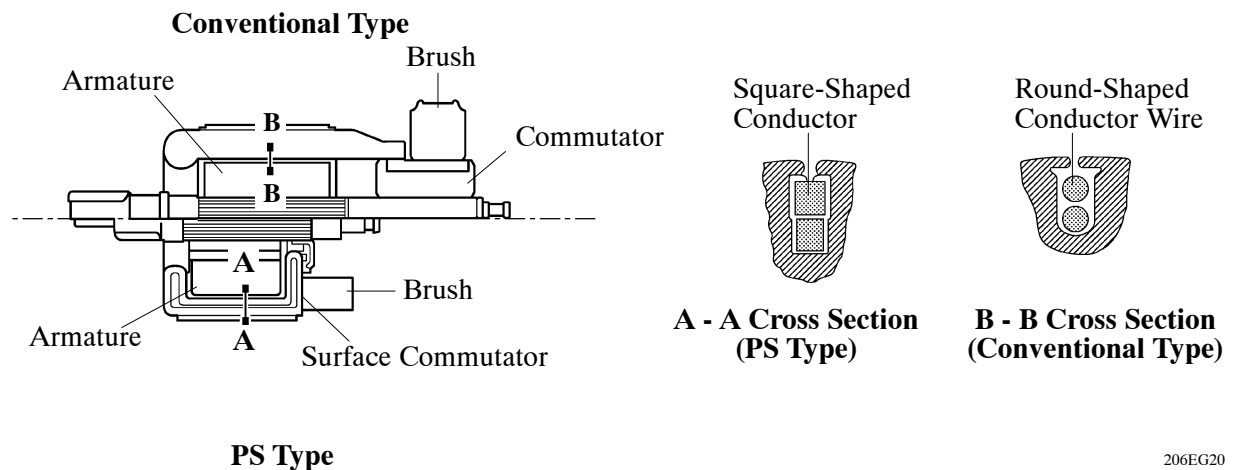


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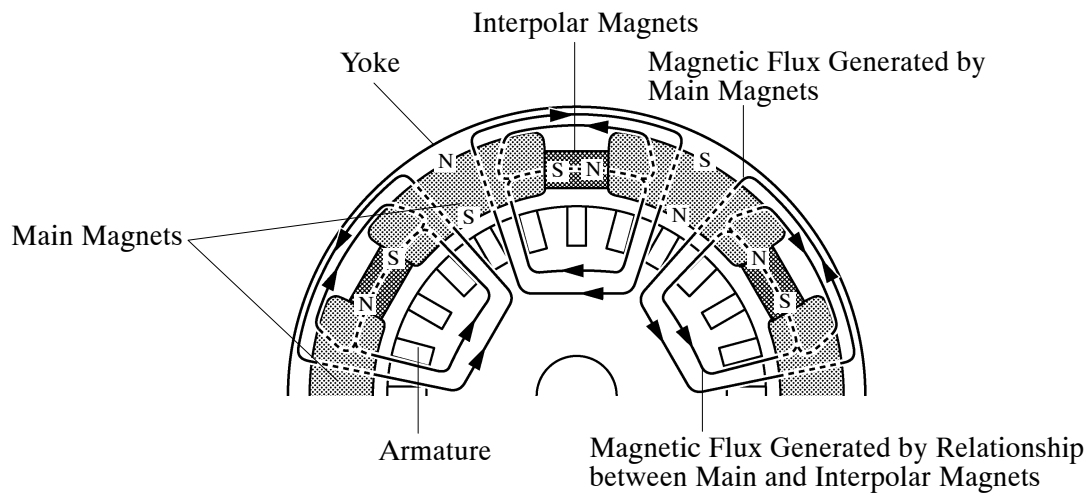
Construction

- Instead of the round-shaped conductor wires used in the conventional starter, the PS type starter uses square-shaped conductors. In this type of construction, square-shaped conductors can achieve the same conditions as those achieved by winding numerous round-shaped conductor wires, but without increasing the mass. As a result, the output torque is increased, and the armature coil is more compact.
- Because the surface of the square-shaped conductors that are used in the armature coil functions as a commutator, the overall length of the PS type starter has been shortened.



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- Instead of the field coils used in the conventional type starter, the PS type starter has adopted two types of permanent magnets: the main magnets and the interpolar magnets. The main and interpolar magnets are arranged alternately inside the yoke, allowing the magnetic flux that is generated between the main and interpolar magnets to be added to the magnetic flux that is generated by the main magnets. In addition to increasing the amount of magnetic flux, this construction shortens the overall length of the yoke.



Cross Section of Yoke Portion

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